Functions & Transformations

ANSWER KEY

Section 1: Domain, Range & Function Types

1 What is the domain of the real-valued function $f(x) = \sqrt{2x - 8}$?

a. $[4, \infty)$ ✓

b. $(4, \infty)$

c. $[2, \infty)$

d. $(-\infty, 4]$

The expression under the radical must be non-negative: $2x - 8 \geq 0 \Rightarrow 2x \geq 8 \Rightarrow x \geq 4$, which is $[4, \infty)$.

2 Which of the following functions is an ODD function?

a. $f(x) = x^3 - 5x$ ✓

b. $f(x) = x^4 + 2x^2$

c. $f(x) = x^2 - 3$

d. $f(x) = x^3 + 1$

An odd function satisfies $f(-x) = -f(x)$. For $f(x) = x^3 - 5x$:
$f(-x) = (-x)^3 - 5(-x) = -x^3 + 5x = -(x^3 - 5x) = -f(x)$.

3 What is the vertical asymptote of the rational function $f(x) = \frac{3x+1}{2x-6}$?

a. $x = 3$ ✓

b. $x = 2$

c. $x = -3$

d. $y = \frac{3}{2}$

Set denominator equal to zero: $2x - 6 = 0 \Rightarrow 2x = 6 \Rightarrow x = 3$.

Section 2: Composite Functions & Inverse Functions

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4 If $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, what is $(f \circ g)(3)$?

a. 19 ✓

b. 17

c. 23

d. 15

First evaluate $g(3) = 3^2 - 1 = 9 - 1 = 8$. Then $f(g(3)) = f(8) = 2(8) + 3 = 16 + 3 = 19$.

5 What is the inverse function $f^{-1}(x)$ for $f(x) = \frac{3x - 5}{2}$?

a. $\frac{2x + 5}{3}$ ✓

b. $\frac{2x - 5}{3}$

c. $\frac{3x + 5}{2}$

d. $\frac{x + 5}{6}$

Set $y = \frac{3x - 5}{2} \Rightarrow 2y = 3x - 5 \Rightarrow 3x = 2y + 5 \Rightarrow x = \frac{2y + 5}{3}$. Swap variables: $f^{-1}(x) = \frac{2x + 5}{3}$.

6 If $f(x) = x^3 - 2$, what is the value of $f^{-1}(25)$?

a. 3 ✓

b. 2

c. 4

d. 5

Set $x^3 - 2 = 25 \Rightarrow x^3 = 27 \Rightarrow x = 3$.

Section 3: Geometric Transformations of Graphs

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7 How is the graph of $g(x) = (x - 3)^2 + 4$ obtained from the parent function $f(x) = x^2$?

a. **Shifted 3 units right and 4 units up** ✓

b. Shifted 3 units left and 4 units up

c. Shifted 3 units right and 4 units down

d. Shifted 3 units left and 4 units down

Inside $(x - 3)$ shifts right by 3 units; $+4$ outside shifts up by 4 units.

8 The graph of $y = f(x)$ is reflected across the x-axis and shifted down by 5 units. What is the equation of the resulting graph?

a. $y = -f(x) - 5$ ✓

b. $y = f(-x) - 5$

c. $y = -f(x - 5)$

d. $y = -f(x) + 5$

Reflection across the x-axis gives $-f(x)$. Shifting down 5 units gives $-f(x) - 5$.

9 What is the vertex of the parabola $f(x) = -2(x + 1)^2 + 7$?

a. $(-1, 7)$ ✓

b. $(1, 7)$

c. $(-1, -7)$

d. $(1, -7)$

In vertex form $y = a(x - h)^2 + k$, the vertex is $(h, k) = (-1, 7)$.

Section 4: Piecewise & Absolute Value Functions

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10 Given the piecewise function $f(x) = 2x + 1$ for $x < 2$, and $f(x) = x^2 - 3$ for $x \geq 2$, evaluate $f(2) + f(0)$.

a. 2 ✓

b. 1

c. 3

d. 4

For $x = 2$, use second rule: $f(2) = 2^2 - 3 = 1$. For $x = 0$, use first rule: $f(0) = 2(0) + 1 = 1$. Sum $= 1 + 1 = 2$.

11 What is the range of the absolute value function $f(x) = |x - 4| + 3$?

a. $[3, \infty)$ ✓

b. $[4, \infty)$

c. $(-\infty, 3]$

d. $[0, \infty)$

Since $|x - 4| \geq 0$, the minimum value occurs at $x = 4$ where $f(4) = 3$. Thus the range is $[3, \infty)$.

12 For what value of x does the function $f(x) = -|x + 2| + 5$ attain its maximum value?

a. -2 ✓

b. 2

c. 5

d. 0

The term $-|x + 2| \leq 0$ is maximized when $|x + 2| = 0 \Rightarrow x = -2$.

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